

MATH 3115 MSc & SED COMPLEX ANALYSIS

Lecture 8: ANALYTIC FUNCTIONS

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Lesson Outcome

By the end of this lesson you should be able to:

- Define Fundamental theorem of Algebra .
- Describe Cauchy-Riemann equations
- Describe Analytic function
- Solve problems involving Harmonic functions

Fundamental theorem of Algebra

Fundamental theorem of Algebra

The fundamental theorem of Algebra states that every non-constant polynomial equation with complex coefficients has at least one complex root.

Moreover, if the degree of the polynomial is n , it will have exactly n complex roots, counting multiplicities.

Example 1

Consider the quadratic equation $x^2 + 2x + 5 = 0$. Find the roots of the polynomials.

Fundamental theorem of Algebra

Solution to example 1

The quadratic formula to solve for x is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Since $a = 1$, $b = 2$, and $c = 5$ then into the formula we have

$$x = \frac{-2 \pm \sqrt{2^2 - 4(1)(5)}}{2(1)}$$

Simplifying the expression inside the square root:

$$x = \frac{-2 \pm \sqrt{4 - 20}}{2}$$

$$x = \frac{-2 \pm \sqrt{-16}}{2}$$

Fundamental theorem of Algebra

Solution to Example 1

Since we have a negative number under the square root, we get an imaginary r

$$\sqrt{-16} = 4i$$

Now substitute this back into the equation:

$$x = \frac{-2 \pm 4i}{2}$$

Now, divide each term by 2 :

$$x = \frac{-2}{2} \pm \frac{4i}{2}$$
$$x = -1 \pm 2i$$

The two roots are:

$$x = -1 + 2i \quad \text{and} \quad x = -1 - 2i$$

Example 2

Consider the cubic equation: $f(z) = z^3 - 1 = 0$. Find the roots of the polynomial

Solution to example 2

Factoring $z^3 - 1$ using the difference of cubes formula we get:

$$z^3 - 1 = (z - 1)(z^2 + z + 1)$$

So the equation becomes:

$$(z - 1)(z^2 + z + 1) = 0$$

Fundamental theorem of Algebra

Solution to example 2 CONT'D

1. For $z - 1 = 0$, we have $z = 1$. Thus our first root.
2. For $z^2 + z + 1 = 0$: We now solve the quadratic equation $z^2 + z + 1 = 0$ using the quadratic formula: $z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
For $a = 1, b = 1$, and $c = 1$, we have:

$$\begin{aligned} z &= \frac{-1 \pm \sqrt{1^2 - 4(1)(1)}}{2(1)} \\ &= \frac{-1 \pm \sqrt{1 - 4}}{2} \\ &= \frac{-1 \pm \sqrt{-3}}{2} \end{aligned}$$

Since $\sqrt{-3} = i\sqrt{3}$, Then we have:

$$z = \frac{-1 + i\sqrt{3}}{2} \quad \text{and} \quad z = \frac{-1 - i\sqrt{3}}{2}$$

Fundamental theorem of Algebra

Solution to example 2 CONT'D

Therefore, the three roots of the equation $z^3 - 1 = 0$ are:

$$z = 1, \quad z = \frac{-1 + i\sqrt{3}}{2}, \quad z = \frac{-1 - i\sqrt{3}}{2}$$

ANALYTIC FUNCTIONS

Cauchy-Riemann equations definition

Cauchy-Riemann equations are a set of conditions that a function must satisfy in order to be complex differentiable (holomorphic) at a point.

Cauchy-Riemann equations CONT'D

If a complex function $f(z)$ is written as:

$$f(z) = u(x, y) + iv(x, y)$$

where:

- $u(x, y)$ is the real part of the function,
- $v(x, y)$ is the imaginary part of the function,
- and $z = x + iy$, where x and y are real numbers (the real and imaginary parts of z), then
- The Cauchy-Riemann equations say that for the function to be differentiable (holomorphic) at a point, the following must be true:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Example

Consider the complex function: $f(z) = z^2 = (x + iy)^2$.
Verify whether $f(z)$ satisfy the Cauchy-Riemann equations

Solution

- $f(z) = (x + iy)^2 = x^2 - y^2 + 2ixy$.
- Real part $u(x, y) = x^2 - y^2$
- and the imaginary part $v(x, y) = 2xy$.
- By Cauchy-Riemann equations
- $\frac{\partial u}{\partial x} = 2x$, and $\frac{\partial v}{\partial y} = 2x$,
- $\frac{\partial u}{\partial y} = -2y$, and $\frac{\partial v}{\partial x} = 2y$.
- Since the Cauchy-Riemann equations are satisfied then $f(z) = z^2$ is holomorphic

Analytic function definition

A function $f(z)$ is analytic at a point z_0 if there exists a neighborhood around z_0 such that $f(z)$ is holomorphic (complex differentiable) at all points in this neighborhood.

Example

Consider the complex function:

$$f(z) = z^2 = x^2 - y^2 + 2ixy$$

Verify whether $f(z)$ is analytic in a complex plane.

solution

$$f(z) = (x + iy)^2 = x^2 - y^2 + 2ixy$$

- The real part $u(x, y) = x^2 - y^2$
- and the imaginary part $v(x, y) = 2xy$
- Since the Cauchy-Riemann equations for $u(x, y)$ and $v(x, y)$ are: 1.

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad 2. \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Compute the partial derivatives of $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$:

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial u}{\partial y} = -2y$$

$$\frac{\partial v}{\partial x} = 2y, \quad \frac{\partial v}{\partial y} = 2x$$

1. $\frac{\partial u}{\partial x} = 2x$ and $\frac{\partial v}{\partial y} = 2x$, so the first equation is satisfied.
2. $\frac{\partial u}{\partial y} = -2y$ and $-\frac{\partial v}{\partial x} = -2y$, so the second equation is satisfied.

Since both Cauchy-Riemann equations hold, the function $f(z) = z^2$ is analytic.

Definition 1

A function is harmonic if both its real and imaginary parts satisfy the Laplace equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Example

Consider the complex function

$f(z) = z^2 = x^2 - y^2 + 2ixy$. Determine whether the real part and imaginary of $f(z)$ is harmonic.

Solution

For $u(x, y) = x^2 - y^2$:

First, Lets compute the second partial derivatives of $u(x, y)$:

$$\frac{\partial^2 u}{\partial x^2} = 2, \quad \frac{\partial^2 u}{\partial y^2} = -2$$

Checking the Laplace equation for $u(x, y)$:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 + (-2) = 0$$

Since the Laplace equation is satisfied, $u(x, y) = x^2 - y^2$ is harmonic.

For $v(x, y) = 2xy$: Now, compute the second partial derivatives of $v(x, y)$:

$$\frac{\partial^2 v}{\partial x^2} = 0, \quad \frac{\partial^2 v}{\partial y^2} = 0$$

Solution CONT'D

Check the Laplace equation for $v(x, y)$:

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 + 0 = 0$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Since the Laplace equation is satisfied, $v(x, y) = 2xy$ is also harmonic.

THE END